

PRUNE-OPD: Efficient and Reliable On-Policy Distillation for Long-Horizon Reasoning

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Project Repo: <https://github.com/yangzhch6/Prune-OPD>

Abstract

On-policy distillation (OPD) leverages dense teacher rewards to enhance reasoning models. However, scaling OPD to long-horizon tasks exposes a critical flaw: as the student’s generated prefix inevitably diverges from the teacher’s thought process, the teacher’s dense reward loses local exploitability. Continuing to generate and evaluate tokens on these “drifted” trajectories not only degrades reward quality but also incurs massive computational waste. To address this, we introduce **Prune-OPD**, a framework that dynamically aligns training budgets with supervision quality. By continuously monitoring the local compatibility between student and teacher predictions (e.g., via top- k overlap), Prune-OPD detects prefix-drift events in real time. Upon detecting severe drift, it monotonically down-weights subsequent unreliable rewards and triggers dynamic rollout truncation. This allows the training process to halt futile generation and reallocate compute strictly to reliable teacher supervision. Across diverse teacher-student combinations, Prune-OPD consistently aligns computation with supervision reliability. When prefix drift makes dense teacher rewards unreliable, it reduces training time by 37.6%–68.0% while preserving, and often improving, performance on challenging benchmarks (AMC, AIME, HMMT). When student-teacher compatibility remains high, it automatically preserves long-context supervision by expanding the training window. These results suggest that Prune-OPD improves OPD not by blindly shortening rollouts, but by reallocating computation toward locally exploitable teacher rewards.

1 Introduction

On-policy distillation (OPD) has become a central technique for post-training large language models. Recent systems like Qwen3 [36], MiMo [34], and GLM-5 [45] adopt OPD-style dense supervision in post-training pipelines. Thinking Machines Lab [24] reports that an OPD recipe can reproduce strong reasoning gains at substantially lower compute than outcome-reward RL. This makes OPD an attractive complement to supervised fine-tuning and verifiable RL: it supplies a token-level learning signal on long reasoning traces, rather than waiting for a sparse final-answer reward.

OPD’s appeal lies in its on-policy nature. Off-policy distillation trains the student on fixed teacher-generated sequences, which induces the exposure-bias problem: at inference time, the student must act under prefixes that were not produced by the teacher or reference policy [2]. OPD instead samples rollouts from the current student and evaluates the teacher on the same student-visited prefixes. This design has also motivated recent self-distillation methods, where a model acts as its own teacher under privileged information or feedback and improves from its own on-policy experience [14, 47, 30]. In long-horizon mathematical reasoning, this distinction matters: a single solution can contain thousands of tokens, so even small train-inference mismatches can compound across the trajectory.

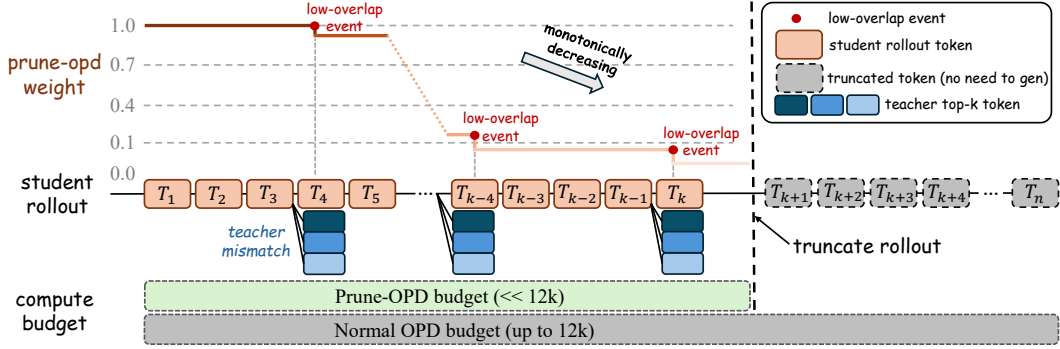


Figure 1: Conceptual overview of PRUNE-OPD. PRUNE-OPD monitors local student-teacher compatibility along the student rollout, monotonically attenuates OPD rewards after low-overlap drift events, and truncates the response once reliable teacher supervision is exhausted.

However, the same on-policy design creates a new reliability problem. The teacher is queried not only on prefixes where its local distribution offers useful corrections, but also on prefixes that have already drifted away from the teacher’s reasoning process. Recent OPD dynamics analysis points out that OPD success depends on local thinking-pattern compatibility and that reward quality can deteriorate with trajectory depth [22]. We take this diagnosis as a design constraint: long-horizon OPD should not treat every generated token as equally worth supervising.

The missing piece is online reliability control. Current OPD recipes usually choose a rollout budget before generation and then apply dense teacher rewards uniformly along the sampled response. This creates a bad tradeoff. A short fixed budget can discard useful long reasoning, while a long fixed budget spends substantial compute generating and scoring suffixes whose teacher signal may no longer be locally exploitable. In practice, a single rollout can contain both regimes: an early prefix where student and teacher still share plausible next-token support, followed by a suffix where the student has committed to reasoning choices the teacher would not have taken. What OPD needs is therefore not simply more or less dense reward, but a way to decide *where* the dense reward is reliable enough to train on.

We propose PRUNE-OPD, a reliability-aware modifier for long-horizon OPD that makes this decision during training. Rather than filtering whole responses or imposing a fixed response length, PRUNE-OPD asks a position-level question: under the same student prefix, do the student and teacher still share a high-probability candidate region? The primary signal is per-position top- k overlap ratio, with teacher top- p /top- k action acceptance as a stricter variant. When the selected compatibility metric falls below its threshold, PRUNE-OPD records a prefix-drift event. Cumulative drift events define a monotone reliability weight that attenuates later OPD rewards, and the same raw reliability signal defines an effective response length for dynamic rollout control. The intended shift is from fixed-budget OPD to reliability-budgeted OPD: dense supervision should be allocated where the teacher remains actionable on the student’s trajectory.

We conduct extensive experiments to evaluate whether reliability-aware pruning improves long-horizon OPD in practice. The results validate the effectiveness of PRUNE-OPD: it reduces training time by 37.6%–68.0% when dense supervision becomes unreliable, while preserving or improving downstream accuracy. Additional runs show that the controller keeps long-context supervision when the reliability signal remains high, supporting the claim that PRUNE-OPD reallocates computation rather than blindly shortening rollouts. In summary, our contributions are:

- We formulate long-horizon OPD as a reliability-allocation problem on student-visited prefixes, connecting trajectory-depth degradation to local compatibility and reward exploitability.
- We propose PRUNE-OPD, which converts cumulative prefix drift into monotone OPD reward attenuation and a dynamic response-length controller, without changing the baseline OPD path when disabled.
- We provide pilot evidence that reliability-aware pruning can reduce OPD training time by 37.6%–68.0% when prefix drift limits reliable supervision while preserving benchmark performance, and we specify fixed-length pruning controls to test whether the gains come from reliability rather than generic shortening.

2 Background and Problem Setup

2.1 On-Policy Distillation

Let x denote an input prompt and $y = (y_1, \dots, y_T)$ a response sampled autoregressively from a student policy π_θ . At position t , write $y_{<t} = (y_1, \dots, y_{t-1})$ for the prefix and define the student and teacher next-token distributions as:

$$p_t(v) \triangleq \pi_\theta(v \mid x, y_{<t}), \quad q_t(v) \triangleq \pi_T(v \mid x, y_{<t}), \quad (1)$$

over vocabulary tokens $v \in \mathcal{V}$. OPD computes supervision on trajectories sampled from the current student, usually by minimizing a reverse-KL objective over student-generated trajectories [9, 1, 22]:

$$\mathcal{L}_{\text{OPD}}(\theta) = \mathbb{E}_{x, y \sim \pi_\theta(\cdot \mid x)} \left[\sum_{t=1}^T D_{\text{KL}}(p_t \parallel q_t) \right]. \quad (2)$$

This token-level decomposition drives OPD’s sample efficiency: every position provides a dense teacher signal instead of waiting for a final-answer reward. It also exposes the central assumption: q_t should be a useful local target on the student-visited state $(x, y_{<t})$. In practice, we approximate this reward on the student’s top- k candidate tokens and obtain a per-position reward vector $r_{t,\cdot}$. Because rewards are already localized by position, a reliability modifier can simply multiply all rewards at position t by a scalar weight before the policy loss. This is the interface used by PRUNE-OPD: reliable prefixes keep their OPD rewards, while drifted prefixes receive attenuated rewards.

2.2 Dynamic Metrics

Following the OPD mechanism analysis [22], let $S_t^{(p)} = \text{TopK}(p_t, k)$ and $S_t^{(q)} = \text{TopK}(q_t, k)$. The overlap ratio proposed in that work is

$$\mathcal{M}_{\text{overlap}} = \mathbb{E}_t \left[\frac{|S_t^{(p)} \cap S_t^{(q)}|}{k} \right], \quad (3)$$

which measures whether the student and teacher assign high probability to a shared candidate region. The entropy gap,

$$\Delta H_t = |H(q_t) - H(p_t)|, \quad (4)$$

tracks whether the two models have similar uncertainty on the same visited state. These metrics are usually reported as diagnostics after training. PRUNE-OPD turns them into online control signals. The current primary metric is the per-position overlap ratio, which measures whether student and teacher still share a high-probability candidate region on the same student prefix. We also evaluate a stricter action-level variant that asks whether the sampled student token is accepted by the teacher’s observable top- p /top- k region.

2.3 Long-Horizon Reliability Problem

The reward at position t depends on $q_t(v) = \pi_T(v \mid x, y_{<t})$, where the prefix $y_{<t}$ was produced by the student. As t grows, the prefix can drift away from trajectories the teacher would naturally generate. Prior OPD analysis reports suffix-to-prefix instability and decreasing teacher-continuation advantage as the student prefix depth increases [22]. Thus the same dense reward that helps on moderate traces can become unreliable on long traces. This motivates position-level reliability control. We estimate whether each student prefix remains teacher-compatible, then allocate dense supervision to reliable prefixes and attenuate rewards after prefix drift.

3 PRUNE-OPD

PRUNE-OPD converts local student-teacher compatibility into a position-wise weight on OPD rewards. The method is designed as a drop-in modifier for OPD: it does not alter how the teacher reward is computed, but changes how much each visited position should trust that reward. It has three parts: a per-position compatibility metric, a cumulative reliability weight, and an optional dynamic response-length controller.

3.1 Compatibility Metric

At position τ , our goal is to decide whether the teacher still provides locally exploitable supervision on the student’s current trajectory. The student and teacher are therefore evaluated on the same student-generated prefix $s_\tau = (x, y_{<\tau})$.

We first restrict the comparison to each model’s high-confidence candidate tokens. Let

$$\mathcal{K}_\tau^S = \text{TopK}(\pi_\theta(\cdot \mid s_\tau), k), \quad \mathcal{K}_\tau^T = \text{TopK}(\pi_T(\cdot \mid s_\tau), k) \quad (5)$$

be the returned top- k candidate sets. The top- k size is an experimental hyperparameter. Then, we measure local compatibility with the overlap ratio

$$\mathcal{O}_\tau = \frac{|\mathcal{K}_\tau^S \cap \mathcal{K}_\tau^T|}{k}. \quad (6)$$

Low overlap means that the teacher’s dense reward is being computed under prefixes where its preferred next-token support has separated from the student’s support. Accordingly, we define the prefix-drift event as

$$B_\tau = \mathbf{1}[\mathcal{O}_\tau < \gamma], \quad (7)$$

where γ is the compatibility threshold. These drift events are later accumulated to attenuate subsequent OPD rewards and define the effective response length for dynamic rollout control.

This metric follows the OPD mechanism analysis directly: successful OPD is associated with increasing high-probability overlap, while failing runs show stagnant or unstable overlap on student-visited states [22]. It measures whether the student and teacher still share a plausible local action space. Low overlap means that the teacher’s dense reward is being computed in a region where its preferred next-token support has separated from the student’s support. We also design an alternative top- p action-acceptance metric, its definition is given in Appendix A.10.

3.2 Cumulative Reliability and Loss Weighting

Bad events are accumulated over the response:

$$C_\tau = \sum_{i=1}^{\tau} B_i. \quad (8)$$

The raw reliability weight is a clipped linear decay,

$$R_\tau = \text{clip}(1 - w_{\text{drop}} C_\tau, 0, 1), \quad (9)$$

where $w_{\text{drop}} \geq 0$ controls the penalty per compatibility failure. The raw weight is monotone non-increasing along each valid response. Padding positions are forced to zero.

The actual loss weight adds a base floor:

$$L_\tau = R_\tau + w_{\text{base}} \quad (10)$$

for valid response positions, and $L_\tau = 0$ for padding. The scaled OPD reward is

$$\tilde{r}_{\tau,j} = L_\tau \cdot r_{\tau,j}. \quad (11)$$

The base weight has two practical roles. First, it prevents PRUNE-OPD from becoming a hard switch when R_τ reaches zero. Second, it lets the method trade off denoising and retention of weak teacher signal. Concrete hyperparameter values are specified in the experimental setup.

3.3 Dynamic Response Budget

PRUNE-OPD can additionally adjust the rollout length during training. Define the effective response length of a sample as

$$E = \sum_{\tau=1}^T \mathbf{1}[R_\tau > \epsilon], \quad (12)$$

for a small tolerance ϵ . This definition deliberately uses R_τ , not L_τ . Therefore a token with $R_\tau = 0$ and $w_{\text{base}} > 0$ can still contribute a small OPD reward, but it is not counted as reliable length.

Algorithm 1 PRUNE-OPD with reliability-aware reward scaling and dynamic response budget

Require: prompt distribution $\mathcal{D}$, student policy π_θ , teacher policy π_T , OPD reward routine, initial response budget M_1

Ensure: updated student policy π_θ and response budget sequence $\{M_t\}$

```
1: for training step  $t = 1, 2, \dots$  do
2:   Sample prompts  $\{x^{(b)}\}_{b=1}^B \sim \mathcal{D}$  and generate student rollouts  $\{y_{1:T_b}^{(b)}\}_{b=1}^B$  with maximum
   length  $M_t$ .
3:   Compute raw OPD reward tensors  $r_{\tau,j}^{(b)}$  on the student-visited prefixes.
4:   for each rollout  $b = 1, \dots, B$  do
5:     Compute the overlap-ratio sequence on student-visited prefixes using Eq. (6).
6:     Mark low-compatibility positions and form the cumulative reliability weights using
     Eqs. (7)–(10).
7:     Scale OPD rewards using Eq. (11) and compute the effective reliable length  $E^{(b)}$  using
     Eq. (12).
8:   end for
9:   Update  $\theta$  with the OPD objective using scaled rewards  $\tilde{r}_{\tau,j}^{(b)}$ .
10:   $h_t \leftarrow B^{-1} \sum_{b=1}^B \mathbf{1}[E^{(b)} \geq M_t - m]$ .
11:  if  $h_t \geq \rho$  then
12:     $M_{t+1} \leftarrow \min(M_t + \Delta, M_{\max})$  and reset the low-hit counter.
13:  else if  $h_t < \rho$  for  $P$  consecutive steps then
14:     $M_{t+1} \leftarrow \max(M_t - \Delta, M_{\min})$  and reset the low-hit counter.
15:  else
16:     $M_{t+1} \leftarrow M_t$ .
17:  end if
18: end for
```

Let M_t be the maximum response length used for rollout generation at training step t . The controller monitors the batch hit ratio: the fraction of samples whose effective reliable length reaches the current limit up to a margin. If the hit ratio is large, the limit expands by a fixed step, capped by $M_{\max}$. If the hit ratio remains small for a patience window, the limit contracts by the same step, bounded below by $M_{\min}$. The initial limit, bounds, update step, margin, hit-ratio threshold, and patience are specified in the experimental setup. Algorithm 1 summarizes the full PRUNE-OPD procedure, including reliability-aware reward scaling and dynamic response-budget adaptation. To keep the training loop readable, the position-wise compatibility and weighting rules refer directly to the definitions above.

4 Experiments

We evaluate whether reliability-aware reward scaling can substantially reduce OPD training cost while preserving downstream reasoning accuracy. Beyond final benchmark accuracy and training time, we measure local student-teacher dynamics on student-visited states, including overlap ratio, top- p acceptance, and effective reliable length.

4.1 Experimental Setup

Training data. We train on DAPO-Math-17K [44], following the OPD setting used in recent reasoning experiments. Training prompts are shuffled before each run. The primary RL outcome reward is final-answer correctness. However, OPD supplies dense token-level distillation rewards.

Models. We evaluate five teacher-student pairs to test the reliability signal across distinct OPD dynamics. DeepSeek-R1-Distill-Qwen-1.5B is distilled from DeepSeek-R1-Distill-Qwen-7B, a larger same-family reasoning model whose higher benchmark score may not imply locally exploitable token-level feedback [10]. The same student is also distilled from JustRL-DeepSeek-1.5B, a same-size RL-improved teacher obtained from DeepSeek-R1-Distill-Qwen-1.5B [11]. Qwen3-1.7B-Base and Qwen3-4B-Base are distilled from Qwen3-4B (Non-thinking), isolating how a thinking-mode shift interacts with student scale within the same model family [36]. Finally, DeepSeek-R1-Distill-Qwen-7B is distilled from Skywork-OR1-7B, providing a high-initial-overlap setting where PRUNE-OPD is

Table 1: Main results for PRUNE-OPD. Accuracy is reported in percent. Time reduction is relative to baseline OPD under the same student-teacher pair.

Method	AMC23	AIME24	AIME25	HMMT24	HMMT25	Time	Red.
<i>DeepSeek-R1-Distill-Qwen-1.5B / JustRL-DeepSeek-1.5B</i>							
OPD	77.5	43.5	32.7	23.5	18.1	6.9h	0%
OPD (Truncate 4k)	77.8	43.1	31.0	22.7	17.9	4.8h	30.4%
PRUNE-OPD (top-p)	77.3	44.0	30.8	23.3	19.8	4.3h	37.7%
PRUNE-OPD (overlap)	78.4	45.2	33.5	22.7	18.8	4.1h	40.6%
<i>DeepSeek-R1-Distill-Qwen-1.5B / DeepSeek-R1-Distill-Qwen-7B</i>							
OPD	66.4	33.5	25.4	14.4	15.0	12.5h	0%
OPD (Truncate 4k)	66.3	31.7	25.8	14.2	14.4	6.6h	47.2%
PRUNE-OPD (top-p)	65.5	29.6	24.2	14.6	14.0	4.8h	61.6%
PRUNE-OPD (overlap)	65.9	33.3	25.8	14.6	15.0	4.0h	68.0%
<i>Qwen3-1.7B-Base / Qwen3-4B (Non-thinking)</i>							
OPD	28.0	7.3	2.9	2.1	0.6	16.5h	0%
OPD (Truncate 4k)	28.6	6.3	2.9	2.3	1.0	12.9h	21.8%
PRUNE-OPD (top-p)	28.4	5.6	2.9	1.9	0.6	11.5h	30.3%
PRUNE-OPD (overlap)	29.1	7.3	4.2	2.9	1.0	10.3h	37.6%
<i>Qwen3-4B-Base / Qwen3-4B (Non-thinking)</i>							
OPD	42.7	14.8	10.6	5.4	2.9	28.5h	0%
OPD (Truncate 4k)	41.9	15.2	13.5	4.8	2.7	22.3h	21.8%
PRUNE-OPD (top-p)	40.5	13.1	12.7	4.8	4.0	20.0h	29.8%
PRUNE-OPD (overlap)	42.7	16.3	14.6	6.7	5.6	13.5h	52.6%

expected to preserve long-context supervision rather than shorten the rollout. Together, these combinations test the same claim under different conditions: the controller reduces computation when reliability decays, while preserving dense supervision when compatibility remains high.

Baselines. We compare baseline OPD, fixed-length truncation baselines, and PRUNE-OPD variants. Fixed-length OPD uses the same distillation objective as baseline OPD but clamps the maximum response length to a fixed budget. Random pruning matches either the retained token fraction or retained OPD reward mass of PRUNE-OPD while choosing pruned positions independently of student-teacher compatibility. These controls test whether the benefit comes from the student-teacher compatibility signal rather than from generic shorter rollouts or lower reward density. The main PRUNE-OPD configuration uses overlap ratio with threshold $\gamma = 0.7$. We also report a teacher top- p action-acceptance variant with $p = 0.95$. Both variants use top- $k = 16$, $w_{\text{drop}} = 0.01$, $w_{\text{base}} = 0.5$, and dynamic-length hit ratio 0.1. Unless otherwise stated, dynamic response length uses initial 2048, min 1024, max 12288, update step 100, margin 100, and shrink patience 3.

Evaluation. We evaluate on AMC23, AIME24, AIME25, HMMT24, and HMMT25. The primary metric is pass@1 accuracy under a fixed evaluation protocol. We also report relative training time reduction, response-length statistics, overlap ratio, top- p acceptance rate, and effective reliable length. Training time is measured relative to baseline OPD under the same student-teacher pair.

4.2 Main Results

The main result is a consistent efficiency gain without meaningful degradation in downstream accuracy. Across the main low-compatibility runs, the overlap-ratio variant reduces training time by 35.7%, 68.0%, 37.6%, and 52.6%, while benchmark accuracy remains close to the corresponding OPD baseline. Accuracy differences are mixed across individual benchmarks and should be interpreted as secondary: PRUNE-OPD slightly improves several AIME/HMMT scores, but also decreases a few entries such as AIME24 under the DeepSeek-R1-Distill-Qwen-7B teacher and HMMT24 under the JustRL-DeepSeek-1.5B teacher. This pattern is consistent with the role of PRUNE-OPD as a reliability-aware efficiency mechanism rather than a standalone accuracy-improvement method. The top- p action-acceptance variant is generally more conservative and often weaker than overlap,

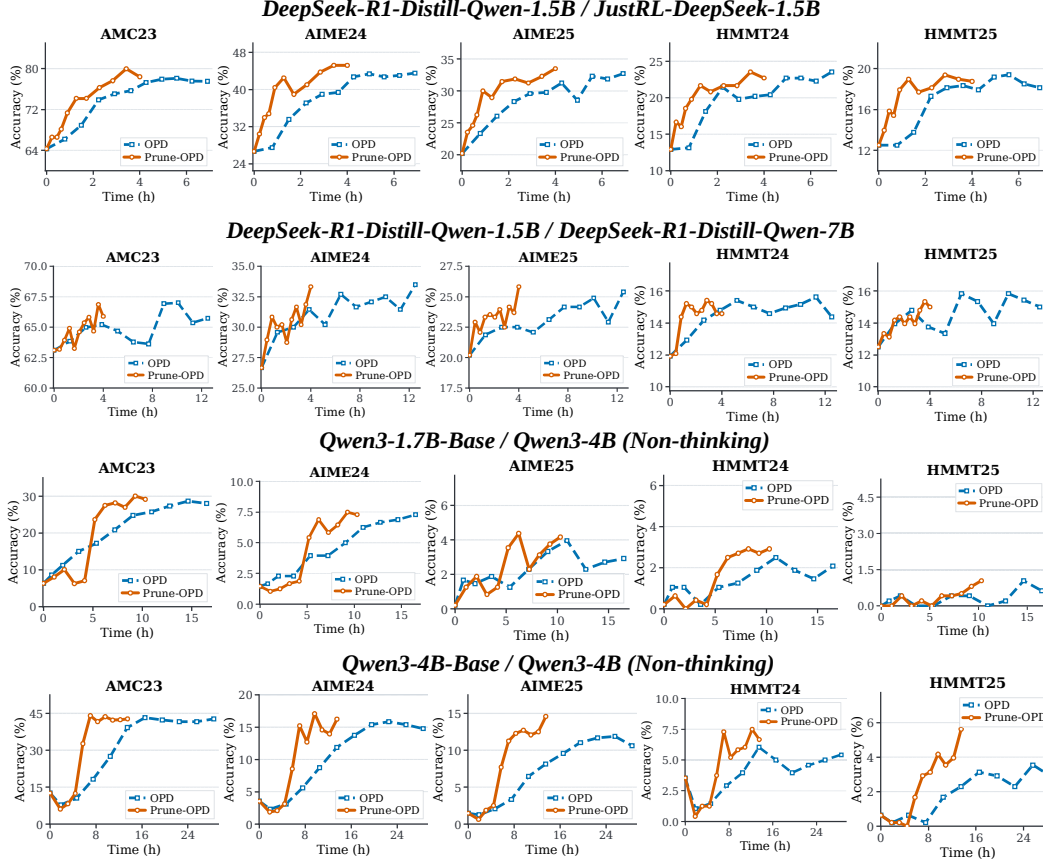


Figure 2: Accuracy over wall-clock time for the 4 DeepSeek student-teacher pairs. Each panel uses wall-clock time as the x-axis and benchmark accuracy as the y-axis, comparing OPD and PRUNE-OPD. A successful curve should match or exceed OPD accuracy while reaching comparable checkpoints earlier in time.

suggesting that exact sampled-token acceptability can be too strict for reasoning traces, whereas candidate-space overlap better captures whether the teacher still supplies exploitable local supervision.

4.3 Adaptive Behavior Under High Compatibility

This setting tests whether PRUNE-OPD uniformly shortens every run. When the teacher and student have high top- k overlap, the effective-length hit ratio is quickly satisfied and the dynamic controller opens the training window to 12288 tokens. The resulting training time and accuracy are nearly unchanged from OPD. Figure 3 should show the corresponding mechanism from three views: the overlap ratio remains high, the dynamic response budget tracks the long reliable context rather than forcing early truncation, and the AMC23 learning curve compares whether PRUNE-OPD matches OPD-level accuracy while avoiding the rigidity of fixed truncation. This supports the intended interpretation that PRUNE-OPD is an adaptive reliability controller, not a hard length penalty.

4.4 Training Dynamics

Figure 4 examines whether PRUNE-OPD changes the learning dynamics or simply removes rollout computation. For the JustRL-DeepSeek-1.5B teacher, the benchmark-wise step-accuracy curves should show that PRUNE-OPD follows the same learning trajectory as OPD on AMC23, AIME24, AIME25, HMMT24, and HMMT25, rather than achieving speed by sacrificing accuracy. Figure 5 explains where the saved computation comes from: the weight-by-position plot should show suffix reward attenuation after accumulated prefix drift, the effective-length plot should show the dynamic maximum OPD length adapting to the reliable supervision window, and the overlap-ratio curve should confirm that pruning is driven by the measured local compatibility signal.

Table 2: High-compatibility behavior. DeepSeek-R1-Distill-Qwen-7B and Skywork-OR1-7B start with overlap ratio around 0.94.

Method	AMC23	AIME24	AIME25	HMMT24	HMMT25	Time	Red.
<i>DeepSeek-R1-Distill-Qwen-7B / Skywork-OR1-7B</i>							
OPD	88.3	67.1	52.5	33.5	32.1	17.5h	0%
OPD (Truncate 4k)	87.2	66.0	48.8	31.0	30.4	11.3h	35.4%
PRUNE-OPD (overlap)	88.9	66.7	52.1	33.1	31.5	17.0h	2.9%

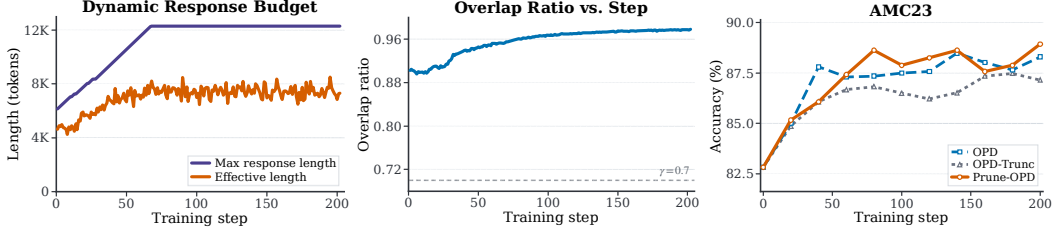


Figure 3: High-compatibility training dynamics for DeepSeek-R1-Distill-Qwen-7B / Skywork-OR1-7B. Left: effective response length and maximum OPD length versus training step. Middle: overlap ratio versus training step. Right: AMC23 accuracy over training, comparing OPD, OPD (Truncate 4k), and PRUNE-OPD.

4.5 Wall-Clock Efficiency

Figure 2 evaluates the core efficiency claim on the wall-clock axis: PRUNE-OPD should reduce elapsed training time without reducing accuracy, and in some cases can slightly improve it. If PRUNE-OPD is working as intended, its curves should lie to the left of OPD at comparable accuracy levels, or above OPD at the same wall-clock time. This presentation directly tests whether PRUNE-OPD reaches OPD-level, and occasionally better, accuracy with less elapsed training time across both DeepSeek teacher-student combinations. Together with the dynamics in Figures 4 and 5, these results show that: reliability-aware suffix down-weighting shortens unproductive rollout computation while preserving the useful prefix supervision that drives final performance.

Table 3: Overlap-threshold ablation on DeepSeek-R1-Distill-Qwen-1.5B / JustRL-DeepSeek-1.5B.

Threshold γ	AMC23	AIME24	AIME25	HMMT24	HMMT25	Time	Red.
0.6	78.9	45.4	33.1	22.0	18.8	4.8h	30.4%
0.7	78.4	45.2	33.5	22.7	18.8	4.1h	40.6%
0.8	77.8	44.5	31.5	21.9	18.0	3.1h	55.1%
0.9	76.8	43.9	30.8	21.8	17.9	2.4h	65.2%

4.6 Ablation Study

OPD with simple truncation. We include OPD (Truncate 4k) as a fixed-budget baseline in Tables 1 and 2. This baseline uses the same OPD objective but shortens every rollout uniformly, without observing student-teacher compatibility. Although simple truncation can reduce cost and may be effective for some teacher-student pairs, it is not a reliable substitute for PRUNE-OPD. In particular, for DeepSeek-R1-Distill-Qwen-1.5B / JustRL-DeepSeek-1.5B and DeepSeek-R1-Distill-Qwen-7B / Skywork-OR1-7B, truncating at a fixed 4K budget degrades performance because it discards useful long-prefix supervision. This failure mode highlights the limitation of a rigid length rule: it cannot distinguish drifted suffixes from long but still compatible reasoning trajectories.

Pruning metric: overlap ratio vs. top- p acceptance. The metric ablation is reported in Table 1. Across the completed runs, overlap ratio is the stronger default: it gives equal or larger time reduction than top- p acceptance while generally preserving accuracy better. We attribute this to its support-level nature: it asks whether student and teacher still share high-probability candidate regions, rather than requiring the sampled student token to fall inside the teacher’s visible nucleus. This makes overlap ratio a smoother reliability signal for long reasoning traces, where multiple locally plausible continuations may remain valid.

Prune-OPD threshold ablation. We ablate the overlap-ratio threshold in Table 3 on DeepSeek-R1-Distill-Qwen-1.5B / JustRL-DeepSeek-1.5B. The completed $\gamma = 0.7$ setting gives a strong

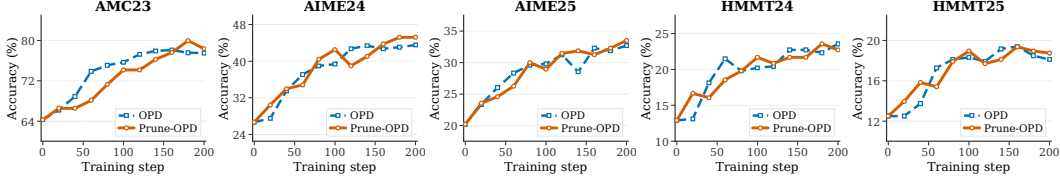


Figure 4: Training-step accuracy dynamics for DeepSeek-R1-Distill-Qwen-1.5B distilled from JustRL-DeepSeek-1.5B. The five panels report benchmark accuracy over training steps on AMC23, AIME24, AIME25, HMMT24, and HMMT25, comparing OPD and PRUNE-OPD.

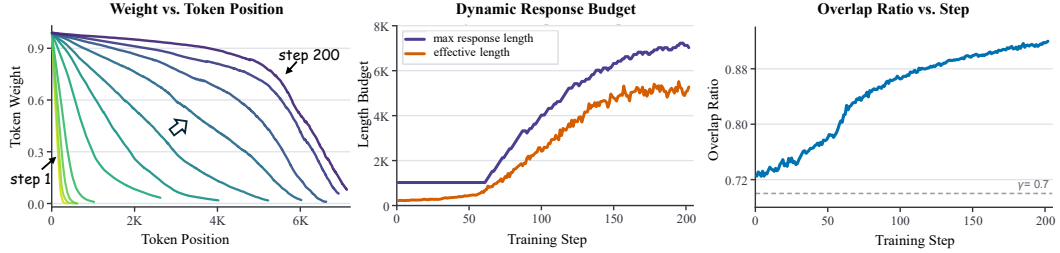


Figure 5: Training-dynamics diagnostics for DeepSeek-R1-Distill-Qwen-1.5B distilled from JustRL-DeepSeek-1.5B. The panels report mean Prune-OPD weight by token position with curves every 20 training steps from 0 to 200; effective response length and maximum OPD length over training; and overlap ratio over training.

operating point: it preserves OPD-level benchmark performance while reducing training time by 35.7%. A lower threshold should be more permissive, producing fewer bad events and longer effective distillation lengths; a higher threshold should prune earlier and may remove useful long reasoning if set too aggressively. The central conclusion is that $\gamma = 0.7$ provides both performance protection and substantial training-efficiency improvement.

5 Related Work

On-policy distillation. MiniLLM formalized OPD for LLMs under a reverse-KL objective optimized by policy gradient, emphasizing the mode-seeking behavior of reverse KL [9]. GKD broadened this view by interpolating between on-policy and off-policy data under multiple divergences [1]. More recent theory frames OPD as dense KL-constrained RL, where the teacher’s per-token log-ratio can be interpreted as an implicit reward and can even be extrapolated beyond the teacher’s standard strength [38]. OPD has since appeared in major reasoning and post-training recipes [36, 24, 45, 34, 20, 17, 15, 7, 38, 8]. It has also been extended to self-distillation and online experiential learning, where a model acts as its own teacher under privileged information, feedback, or sample routing [14, 47, 12, 30, 43, 26, 18, 42, 37, 21, 46, 6, 31]. These works establish OPD as a strong post-training primitive. PRUNE-OPD addresses a complementary reliability problem: even when dense teacher rewards are available on-policy, long rollouts may contain prefixes where the teacher’s local signal is no longer exploitable. More discussions of OPD are provided in Appendix A.5

6 Conclusion

OPD provides dense supervision for long reasoning chains; however, when the student’s reasoning pattern diverges from that of the teacher, the resulting dense rewards become unreliable. PRUNE-OPD treats this as a position-wise reliability problem: it checks whether student and teacher remain locally compatible under shared prefix, turns cumulative compatibility failures into a drift signal, rescales OPD rewards, and controls response length using reliable rather than raw token count. Across diverse distillation settings, overlap-based PRUNE-OPD reduces training time by 37.6%–68.0% in low-compatibility regimes while preserving benchmark performance, and it keeps the long training window open when compatibility remains high. These results show that efficient long-horizon OPD should not be governed by teacher strength or a fixed rollout budget alone. Instead, dense teacher supervision should be allocated according to its local exploitability on the prefixes that the student actually visits.

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A Appendix

A.1 Broader Impact

This research contributes to the development of more efficient and reliable training protocols for LLMs specializing in complex reasoning. By introducing PRUNE-OPD, we provide a mechanism to mitigate the risks of “reward hacking” and training instability inherent in dense on-policy distillation, where models might otherwise learn from low-quality or drifted teacher signals. The significant reduction in training compute (up to 68%) directly addresses the environmental concerns associated with the high energy consumption of LLM post-training. By making sophisticated reasoning models more accessible to train with fewer resources, this work democratizes the ability to develop high-performing AI, potentially accelerating progress in scientific discovery.

Despite these benefits, we recognize that enhancing the reasoning capabilities of LLMs could inadvertently lower the barrier for generating sophisticated misinformation or dual-use content. While PRUNE-OPD focuses on the process of learning rather than the content of the data, the resulting models could be more persuasive or harder to detect if used for deceptive purposes. To mitigate these risks, we emphasize the importance of using PRUNE-OPD in conjunction with safety-aligned teacher models and rigorous verifiable RL frameworks. We are committed to transparency

and reproducibility by providing detailed experimental setups and hyperparameter configurations, enabling the community to audit the reliability of these training dynamics and ensure that reasoning improvements do not come at the cost of model safety or ethical alignment.

A.2 LLM Usage Declaration

This manuscript uses LLMs strictly for the purpose of language editing and textual polishing to enhance presentation quality. We declare that the novel ideas, methodological framework, experimental execution, and data analysis are the original work of the authors. All content modified by AI tools has been carefully reviewed and validated by the authors to ensure accuracy.

A.3 Limitations

PRUNE-OPD relies on top- k local-distribution statistics. The primary overlap metric can miss cases where student and teacher share candidate tokens but rank or score them very differently, while the top- p action-acceptance variant can be too strict for stylistically different but valid reasoning paths. The current decay is linear in cumulative bad events and assumes that local compatibility correlates with reward exploitability; this can fail when a low-overlap prefix later returns to a teacher-compatible state. The implementation also only rescales OPD rewards and does not attach a GRPO fallback to zero-reliability tokens. Finally, the current experiments are limited to math reasoning with DeepSeek, Qwen, and Skywork-style models; agentic and multi-turn settings remain future work.

A.4 Future Work

A natural extension is to combine PRUNE-OPD with GRPO in a reliability-gated objective. Before the reliability weight decays to zero, the model would still optimize the OPD loss and benefit from dense teacher guidance on locally compatible prefixes. After the reliability weight reaches zero, the unreliable suffix would switch to a GRPO loss, allowing the student to continue exploring with outcome-level feedback instead of forcing it to follow a teacher distribution that is no longer locally exploitable. Such a hybrid objective could combine the sample efficiency of teacher-guided distillation with the exploratory benefits of reinforcement learning, and may further improve long-horizon reasoning beyond reliability-aware truncation alone.

A.5 Additional Related Work

Knowledge distillation and off-policy distillation. Knowledge distillation transfers information from a teacher to a student by matching the teacher’s soft output distribution [13]. For autoregressive sequence models, sequence-level distillation trains the student on teacher-generated outputs [19], and this off-policy paradigm underlies many compact language-model distillation methods [27, 16, 32, 8]. Supervised fine-tuning and instruction tuning similarly improve downstream behavior by training on curated responses or demonstrations [5, 28, 33]. The common limitation is train-inference distribution mismatch: the student is optimized on teacher or reference trajectories, but at inference time it must condition on prefixes sampled from its own policy, an exposure-bias problem that compounds over long generations [2]. PRUNE-OPD follows the OPD motivation that distillation should be computed on student-visited states, but asks a further question: once the student is on-policy, which visited states still admit reliable dense teacher supervision?

OPD dynamics and mechanisms. Recent OPD analysis shows that OPD success is governed by thinking-pattern consistency and by whether the teacher contributes new knowledge beyond what the student has already acquired [22]. At the token level, successful OPD exhibits progressive alignment on high-probability tokens: top- k overlap rises, overlap-token advantage improves, and the entropy gap narrows. The same analysis further shows that reward quality degrades with trajectory depth, and that a reward can remain globally correlated with final correctness while failing to provide locally useful gradients around the student’s current policy. PRUNE-OPD directly operationalizes these diagnostic findings. Instead of treating overlap ratio, entropy gap, or teacher acceptability as post-hoc measurements, it turns per-prefix compatibility into an online reliability signal that attenuates future OPD rewards and controls rollout length.

Capacity gap and distillability. A separate line of work studies when a teacher is too strong, too different, or too complex for the student to imitate. Studies have shown that overly capable teachers can hurt student performance [4], and teacher assistants have been introduced to bridge large capacity gaps [25]. Distillation scaling laws identify non-monotonic interactions among teacher quality, student size, and data volume [3]. For reasoning models, small students can struggle to learn from strong reasoners with long chain-of-thought traces, suggesting a learnability gap in distilling complex reasoning behavior [23]. These results caution against assuming that teacher benchmark strength alone determines distillation quality. PRUNE-OPD studies the same issue from an on-policy, position-wise perspective: a teacher may be useful globally but unreliable at specific depths of a student-generated trajectory.

Reasoning RL and long-horizon supervision. Modern reasoning models rely on verifiable RL, long chain-of-thought rollouts, and large mathematical training sets [29, 10, 44, 11, 39, 40, 41, 35]. Long responses create room for search and self-correction, but they also make dense supervision expensive and harder to trust. Prior OPD results identify a response-length sweet spot: short responses provide too little dense signal, while very long responses can trigger late-stage collapse and suffix-to-prefix instability [22]. PRUNE-OPD is designed for this long-horizon setting. It does not impose a fixed response budget; it estimates which prefixes still carry reliable teacher signal and uses that estimate for both reward scaling and dynamic response-length control.

A.6 Reliability Rationale

PRUNE-OPD is motivated by the distinction between globally informative teacher rewards and locally exploitable token-level supervision on student-visited prefixes [22]. Overlap ratio and related compatibility metrics are therefore used online: repeated low-compatibility events are treated as evidence that the student trajectory has drifted into a region where uniform dense OPD rewards are less trustworthy. The method does not assume that every low-overlap token is wrong; it only assumes that cumulative compatibility failure is a useful path-dependent signal for attenuating later suffix rewards.

The monotone reliability decay is a deliberately conservative design. Later tokens can become locally compatible because the teacher adapts to an already drifted prefix, not because the full reasoning trajectory has returned to a teacher-compatible state. Using cumulative decay avoids re-amplifying suffix rewards after repeated drift events, while the base weight w_{base} separates the reliability estimate R_τ from the optimization scale L_τ . This lets PRUNE-OPD down-weight unreliable suffix supervision without turning the method into a brittle hard filter.

A.7 Implementation Details

PRUNE-OPD is implemented as a post-processing step on the OPD reward tensor. The baseline reward tensor has shape $[B, T, k]$, where B is batch size, T is response length, and k is the top- k token dimension. For each valid response position, the method computes the scalar loss weight L_τ and multiplies every candidate reward at that position by the same scalar:

$$\tilde{r}_{b,\tau,j} = L_{b,\tau} r_{b,\tau,j}. \quad (13)$$

Padding positions have $R_\tau = 0$ and $L_\tau = 0$. The advantage estimator remains `token_reward_direct`, so the scaled reward is used directly as the token-level advantage.

The current code path keeps the baseline isolated. When `prune_opd.enable=False`, the original OPD/GRPO behavior is unchanged. When enabled, the main experiments use `overlap_ratio`; `teacher_top_p_accept` is retained as an action-level variant. The teacher top- k size is controlled by `actor_rollout_ref.rollout.log_prob_top_k` and is set to 16 in the current experiments.

A.8 Prompt Template

We use the same DAPO-Math prompt template as the OPD dynamics study [22]. For each math problem, the raw question is inserted into the following instruction.

DAPO-Math Prompt Template

{ Question } Please reason step by step, and put your final answer within `\boxed{ }`.

A.9 Primary Configurations

Table 4 summarizes the primary hyperparameters used across our PRUNE-OPD experiments. Unless otherwise stated, all benchmark performance reported in the paper is Avg@16 accuracy: for each benchmark problem, evaluation uses 16 sampled responses under the fixed evaluation protocol and reports the average correctness across those samples. For high compatibility pair (DeepSeek-R1-Distill-Qwen-7B / Skywork-OR1-7), the init dynamic length is set as 6144; For other pairs, the init dynamic length is set as 1024.

Table 4: Default hyperparameters for PRUNE-OPD.

Item	Value
Training dataset	DAPO-Math-17K
Training temperature	1.0
Teacher temperature	1.0
Rollout number	4
Mini batch size	64
LogProb top- K	16
Top- K strategy	Student Top- K
Reward weight mode	Student probability
Max prompt length	1,024
Max response length	8,192–12,288
Validation max response length	31,744
Learning rate	1×10^{-6}
Training steps	203
KL coefficient	0.0
Loss aggregation	token-mean
Overlap threshold γ	0.7
Top- p threshold p	0.95
w_{drop}	0.01
w_{base}	0.5
Dynamic length init/min/max	1024 / 1024 / 12288
Dynamic length step	100
Hit ratio threshold	0.1
Length margin	100
Shrink patience	3

A.10 Top- p Action-Acceptance Metric

In addition to overlap ratio, we evaluate a stricter action-level compatibility metric. At position τ , the student samples the actual next token y_τ , while the teacher returns top- k tokens and probabilities under the same prefix:

$$\mathcal{K}_\tau = \{v_{\tau,1}, \dots, v_{\tau,k}\}, \quad q_{\tau,1} \geq q_{\tau,2} \geq \dots \geq q_{\tau,k}. \quad (14)$$

Given a top- p threshold p , define the cumulative visible mass

$$S_{\tau,m} = \sum_{j=1}^m q_{\tau,j}. \quad (15)$$

If a smallest m_τ satisfies $S_{\tau,m_\tau} \geq p$, the visible nucleus is

$$\mathcal{A}_\tau(p, k) = \{v_{\tau,1}, \dots, v_{\tau,m_\tau}\}. \quad (16)$$

If the returned top- k mass is still below p , the true nucleus extends beyond the observed set, so we conservatively use $\mathcal{A}_\tau(p, k) = \mathcal{K}_\tau$. The bad event is $B_\tau = \mathbf{1}[y_\tau \notin \mathcal{A}_\tau(p, k)]$. In the reported top- p runs, $p = 0.95$.

A.11 Why PRUNE-OPD Can Improve Accuracy for Low-Overlap Qwen Pairs?

Table 1 shows that the two Qwen3 teacher-student pairs, Qwen3-1.7B-Base / Qwen3-4B (Non-thinking) and Qwen3-4B-Base / Qwen3-4B (Non-thinking), are the clearest cases where PRUNE-OPD improves both training efficiency and benchmark accuracy. We interpret this as a low-overlap

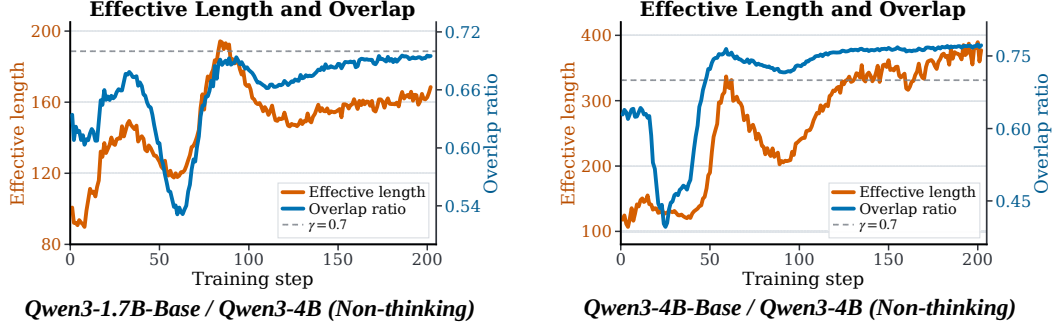


Figure 6: Short effective OPD windows in the low-overlap Qwen3 distillation pairs. For Qwen3-1.7B-Base / Qwen3-4B (Non-thinking) and Qwen3-4B-Base / Qwen3-4B (Non-thinking), low overlap causes PRUNE-OPD to concentrate OPD supervision within a few hundred reliable tokens, whereas the OPD baseline keeps training on responses up to 12,288 tokens.

regime where reliability-aware pruning acts not only as a compute-saving mechanism, but also as a gradient denoising mechanism. As shown in Figure 6, these pairs exhibit low student-teacher overlap, and PRUNE-OPD therefore keeps the effective OPD length at only a few hundred tokens. In contrast, the corresponding OPD baseline uses a maximum response budget of 12,288 tokens.

This mismatch creates a plausible failure mode for unpruned OPD. Only the early prefix contains locally exploitable teacher supervision, while the long suffix contributes a large number of token-level gradients produced on drifted, low-overlap prefixes. Because those suffix tokens vastly outnumber the useful prefix tokens, their gradients can dominate the update even if each individual suffix reward is weak or noisy. PRUNE-OPD removes this imbalance by attenuating and truncating low-reliability suffixes, concentrating optimization on the short prefix where the teacher remains actionable. This explains why these two Qwen3 pairs can gain accuracy in addition to saving time. For the other teacher-student combinations, compatibility is higher or the useful supervision window is already long enough that pruning mainly reduces computation while preserving accuracy, rather than substantially changing the effective training signal.

A.12 Additional Diagnostics

The appendix diagnostics focus on the compatibility signals that motivate PRUNE-OPD. Figure 7 reports how overlap ratio evolves across depth bands under unpruned OPD, and Figure 8 shows how the overlap threshold controls suffix down-weighting and the dynamic response budget.

Depth-wise overlap under OPD. Figure 7 isolates the compatibility dynamics of the unpruned OPD baseline. The shallow 0–1K band acts as a reference for early-prefix supervision, while the later bands test whether teacher supervision remains locally aligned after long student-generated reasoning prefixes. The key observation is that compatibility is depth-dependent: later token-position bands are the regime where overlap is most fragile and where uniform dense rewards are least likely to provide useful local gradients. This supports the central premise of PRUNE-OPD: the inefficiency of long-horizon OPD is not only that suffixes are expensive, but that the suffix rewards are often produced on prefixes where the teacher and student have already drifted apart.

Threshold sensitivity. Figure 8 explains how the reliability threshold changes the intervention. A larger threshold declares compatibility failures earlier, which makes the token-level weight decay sooner and yields a shorter dynamic OPD budget; a smaller threshold is more permissive and preserves longer suffix supervision. The main setting $\gamma = 0.7$ lies between these extremes: it removes a substantial amount of low-reliability suffix computation while avoiding the overly rigid behavior of a hard fixed-length truncation. Thus the threshold primarily controls where PRUNE-OPD sits on the reliability–compute axis, rather than changing the underlying OPD objective.

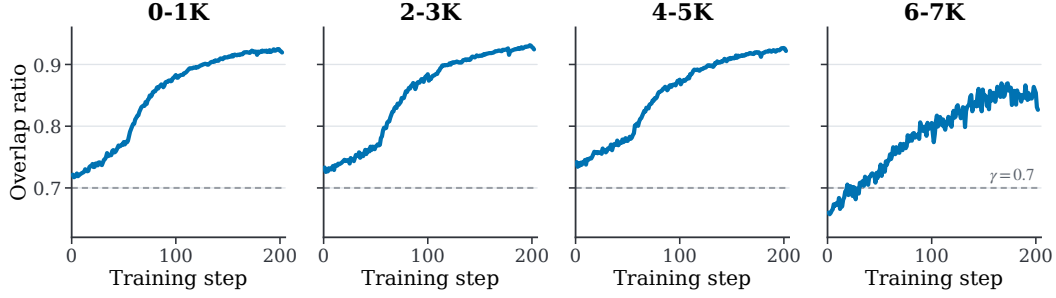


Figure 7: OPD baseline overlap-ratio training dynamics for DeepSeek-R1-Distill-Qwen-1.5B / JustRL-DeepSeek-1.5B. Each panel plots overlap ratio versus training step for a token-position band: 0–1K, 2–3K, 4–6K, and 7–8K. This diagnostic shows how local student-teacher compatibility evolves at different trajectory depths under unpruned OPD.

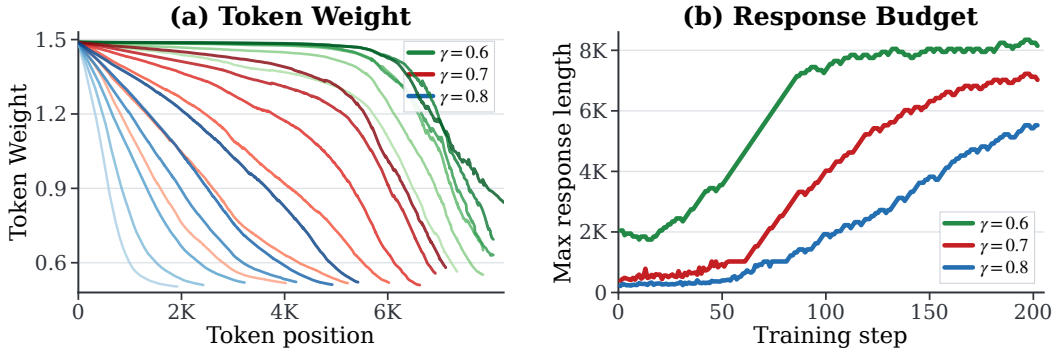


Figure 8: Prune-OPD threshold diagnostics for DeepSeek-R1-Distill-Qwen-1.5B / JustRL-DeepSeek-1.5B. Left: mean Prune-OPD weight as a function of token position under three overlap thresholds, $\gamma = 0.6, 0.7, 0.8$; for each threshold, the curves are taken at training steps 100, 120, 140, 160, 180, and 200. Right: maximum OPD response length over training steps under the same thresholds. Together, these diagnostics show how the reliability threshold changes suffix down-weighting and dynamic rollout budgets.

A.13 Compute Reporting

The dominant compute cost comes of our work from post-training student models with OPD-style teacher scoring and from benchmark evaluation. For each teacher-student pair, OPD, fixed truncation, and PRUNE-OPD are run under the same training stack and matched hardware allocation, so the reported wall-clock comparisons isolate the effect of reliability-aware pruning rather than changes in infrastructure. The main compute evidence is therefore reported as elapsed training time in Tables 1, 2, and 3, and in Figure 2